

Timeformer: Explicit Temporal Conditioning for Traceable Semantic Representations

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Abstract. Words change meaning over time, yet most computational models treat token representations as epoch-independent. We study temporal traceability: the property that the same surface token, queried at different epochs, occupies semantically appropriate neighborhoods in representation space. A model may use epoch information to improve predictive accuracy without exposing a traceable `token@time` representation — and this gap is the focus of the paper.

We address it with Timeformer, a family of Transformer-style encoders organized as a controlled ablation chain. Four variants add temporal capacity in successive steps: a static baseline, a globally time-conditioned encoder, a token-time encoder that conditions each token embedding on the epoch before self-attention is applied, and a memory-augmented extension that consults causal historical prototypes. All variants are trained with masked language modeling on a controlled synthetic benchmark in which the temporal trajectory of each target item is known and evaluation labels are withheld from training.

The main finding is that explicit token-time interaction is the effective mechanism for traceability. The Token-Time Transformer produces a neighborhood drift of -0.614 for drifting subjects across epochs, compared to -0.231 for the static baseline — a difference stable across seeds and visible in class-specific trajectory curves. Probe-based accuracy gains are modest and seed-sensitive, confirming that geometric diagnostics are more appropriate than label recoverability for evaluating traceability. The memory-augmented variant does not produce a robust improvement over token-time interaction; the oracle diagnostic identifies mean-prototype aggregation as the failure point, motivating sense-aware memory mechanisms in future work.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not keep a fixed meaning across time. A term that is associated with one set of events, practices, or communities in an earlier period may later become associated with another. For computational models, this raises a traceability problem: when the same surface token appears in different epochs, we need to inspect not only what the model predicts, but also whether its representation

places that token in the appropriate temporal semantic neighborhood. Without a way to inspect temporal movement in the representation, downstream analyses risk collapsing distinct historical states into a single timeless representation. This limits our ability to audit semantic movement, diagnose temporal generalization, and explain why a model treats the same lexical item differently across periods.

The dominant computational treatment of this problem has been lexical semantic change detection (LSCD), often based on diachronic word embeddings or related distributional representations trained over time-sliced corpora. This line of work has produced important methods and benchmarks for measuring semantic change across periods [3, 6, 7]. However, many approaches compare representations across independently modeled or post-hoc aligned periods. Even when temporal information is available to a model, the resulting representation is not always exposed as a directly queryable object of the form `token@time`. As a result, the temporal behavior of a lexical item may be visible only indirectly through aggregate scores, rather than through the geometry of the representation itself. Furthermore, a model that receives epoch information to improve a prediction objective does not necessarily produce a traceable representation: it may shift its outputs while leaving the geometric neighborhood of the token largely unchanged. Predictive performance and representational traceability are separable properties.

This paper studies this separation directly. We say a representation is *temporally traceable* if the same surface token, queried at different epochs, occupies neighborhoods that reflect its intended temporal context — making it inspectable as a first-class `token@time` object. A model may be temporally predictive without being temporally traceable: the representation geometry may not reflect the token’s trajectory even when the model output does. We therefore ask: can a Transformer-style encoder expose a temporally traceable representation in which the same lexical item, evaluated at different epochs, has measurably different and semantically appropriate neighborhoods?

We address this question with Timeformer, a family of Transformer-style models designed around an explicit temporal interface. The family is organized as an ablation chain: a Static Transformer without time information; an Additive Time-Conditioned Transformer that injects a global epoch vector; a Token-Time Transformer that makes lexical identity interact explicitly with continuous time encodings; and a Memory-Augmented Timeformer that adds causal historical prototype memory. This design lets us isolate which forms of temporal capacity support traceable semantic representations, rather than treating temporal conditioning as a single monolithic architectural choice.

The paper contributes to the literature and state of practice in four ways. First, it formulates temporal semantic traceability as a `token@time` representation problem. Second, it introduces a controlled Timeformer ablation family for comparing global time conditioning, token-time interaction, and historical prototype memory. Third, it evaluates temporal traceability with diagnostics that inspect subject representations directly, including context drift score, contrastive temporal inversion, and probe-based evaluation. Fourth, it shows that

Figure 1. Timeformer ablation chain and token@time interface

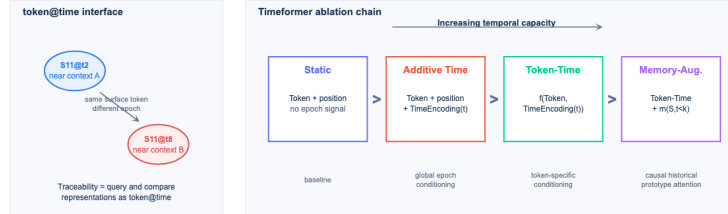


Fig. 1. Conceptual overview of temporal traceability and the Timeformer ablation chain. The same lexical item is inspected as token@time, and model variants add temporal capacity in controlled steps.

token-time interaction is the main positive mechanism in the current setup, while naive mean-pooled historical memory is better understood as a diagnostic extension than as a robust improvement.

The remainder of the paper is organized as follows. Section 2 reviews related work on diachronic representations, LSCD, temporal conditioning, and diagnostic evaluation. Section 3 summarizes the controlled benchmark used as the testbed. Section 4 presents the Timeformer architecture and its ablation variants. Section 5 describes the evaluation protocol. Section 6 reports the empirical results, and Section 7 discusses the implications and limitations of the memory-augmented variant. Section 8 concludes the paper.

2 Related Work

Computational approaches to semantic change are commonly framed as lexical semantic change detection: given language data from different periods, the goal is to determine whether and how the meaning of a target item has changed. Diachronic word embeddings provided one of the most influential formulations of this problem by learning distributional representations for different time slices and comparing them across periods [3]. This family of methods made semantic change operational through geometric movement in embedding space, often after alignment or normalization across independently trained temporal slices. The central intuition remains important for this paper: semantic change should be visible as a change in neighborhood structure. Our focus, however, is not on aligning independently trained spaces. Instead, we ask whether a single Transformer-style encoder can expose a representation that is directly queryable as a lexical item at a specific epoch.

Benchmarks for LSCD have helped standardize evaluation by defining target words, periods, and gold annotations for semantic change. SemEval-2020 Task 1, for example, established an unsupervised LSCD setting with multilingual data and evaluation protocols for binary and graded change detection [6]. Surveys of the field also emphasize the diversity of corpus designs, annotation regimes, and model families used to study semantic change [7]. These resources are essential for evaluating change in natural corpora, but they also inherit the observational complexity of historical language: changes in frequency, genre, domain, and corpus composition may interact with lexical meaning. In this paper, we use a controlled benchmark as a diagnostic testbed rather than as a replacement for natural LSCD benchmarks. The controlled setting gives access to known temporal trajectories and evaluation labels, allowing us to test whether the intended temporal movement is visible in the model representation.

A second relevant line of work concerns temporal conditioning in neural models. Rather than training separate models for each period, a temporally conditioned model can receive time as part of its input or architecture, allowing parameter sharing across epochs while representations or predictions vary with time [1, 5, 4]. For the traceability problem, however, the form of conditioning matters. A global epoch vector may tell the model which period it is processing, but it does not necessarily force an interaction between a particular token identity and that epoch. Timeformer is therefore organized as an ablation chain that separates global time injection from explicit token-time interaction. This distinction is important because the paper evaluates not only whether time improves a prediction, but whether the same token occupies different and interpretable neighborhoods at different epochs.

Finally, this paper is related to diagnostic evaluation and multi-sense representations. Probing methods are often used to ask what information is recoverable from learned representations, but a high probe score does not by itself explain how individual lexical items move through representation space. We therefore combine probe-based evaluation with geometric diagnostics, such as nearest-neighbor context drift, that inspect the representation of the planted subject token directly. Multi-sense representations are also relevant because semantic change may involve the coexistence, competition, or bifurcation of senses rather than a simple shift from one meaning to another [2]. Contextualized models such as ELMo and BERT can assign different representations to the same surface form depending on context, and this property has been used to track sense-level change across periods. The memory-augmented variant studied here uses mean-pooled historical prototypes and is deliberately limited: it tests whether a simple causal memory helps traceability, while also exposing why more sense-aware memory mechanisms are needed in future work.

3 Controlled Benchmark Summary

We evaluate Timeformer on a controlled synthetic benchmark designed to make temporal semantic movement observable. The benchmark is not intended to

replace natural LSCD datasets. Instead, it provides a diagnostic environment in which the temporal trajectory of each target item is known, the source of semantic evidence is explicit, and representation-level behavior can be compared against planted ground truth. Full benchmark construction is treated separately in the companion benchmark paper; here we summarize only the elements needed to understand the Timeformer evaluation.

The corpus uses a strict subject–verb–object (SVO) grammar. This structure isolates subjects as the lexical items with planted temporal trajectories, while verbs and objects serve as noisy contextual evidence. A subject’s meaning at a given epoch is operationalized as the probability that its sentence is generated from one of two latent semantic contexts: $P(A \mid s, t)$ denotes the context-A probability for subject s at epoch t . A subject undergoes semantic change when $P(A \mid s, t)$ shifts across epochs. The two contexts are defined by disjoint sets of verbs and objects. The vocabulary contains 30 subjects, 8 verbs, and 8 objects; context A contains verbs $\{V1, \dots, V4\}$ and objects $\{O1, \dots, O4\}$, while context B contains verbs $\{V5, \dots, V8\}$ and objects $\{O5, \dots, O8\}$.

Subjects are divided into three planted trajectory classes. Stable subjects maintain an approximately constant $P(A \mid s, t)$ across all epochs. Drift subjects follow a monotonic trajectory from near-exclusive context A ($P(A \mid s, t) \approx 1$ at t_0) toward near-exclusive context B ($P(A \mid s, t) \approx 0$ at t_9). Bifurcating subjects begin in a dominant context A state and move toward a mixed regime ($P(A \mid s, t) \approx 0.5$) in which both contexts coexist with roughly equal weight. Full trajectory specifications are given in the companion benchmark paper; here, these classes define the expected direction and magnitude of neighborhood movement used in evaluation. The benchmark contains 10 epochs, t_0 through t_9 , and 10 subjects per trajectory class. This design lets us test whether a model distinguishes stability, directional drift, and movement toward a mixed semantic state.

The observed verb and object are informative but deliberately noisy. For the main corpus, if the planted `true_context` is A, each local marker is sampled from context A with probability $p_{\text{canon}} = 0.75$ and from context B with probability 0.25; the reverse holds when the planted context is B. Thus, local evidence usually points in the intended direction but cannot be used as a deterministic oracle. The `true_context` label is defined by the planted subject–epoch trajectory, not by the observed verb or object, and is used only for evaluation.

The benchmark contains several evaluation conditions. The standard test split uses the same noisy marker process as training. Harder splits include cases in which the verb contradicts the planted context and cases in which both local markers contradict it. An additional ambiguous split is generated with $p_{\text{canon}} = 0.50$, making verb and object markers locally uninformative; in this setting, the subject identity and epoch are the relevant evidence. We also use contrastive evaluations, in which the same subject is presented under alternative epoch labels to test whether the representation inverts accordingly, and continuation evaluations, in which the model is tested on epochs held out from the main training distribution. Both are described in detail in Section 5.

For this paper, the benchmark plays two roles. First, it gives us direct access to target trajectories, so we can ask whether the representation of a subject at t_0 and t_9 occupies the intended semantic neighborhoods. Second, it separates context-local cues from temporal cues, allowing the evaluation to distinguish models that merely exploit observed verbs and objects from models that encode the interaction between lexical identity and epoch. These properties make the benchmark suitable for testing temporal traceability, while also delimiting the scope of the claims: the results are diagnostic evidence in a controlled setting, not a claim of performance on open-domain historical language.

4 Timeformer Architecture

Timeformer is defined as an ablation family rather than as a single model. All variants share the same basic training interface: an input sentence is encoded as [CLS] S V O [SEP], a Transformer encoder produces contextual hidden states, and an MLM head predicts masked tokens. The representation used for traceability diagnostics is h_s , the final hidden state at the subject position. The variants differ only in how epoch information is made available to the encoder and whether the current subject representation can consult causal historical prototypes. Table 1 summarizes the four variants.

Table 1. Model variants in the Timeformer ablation family.

ID	Public name	Temporal signal	Memory
B1	Static Transformer	None	No
B2a	Additive Time-Conditioned Transformer	Global epoch vector	No
B2b	Token-Time Transformer	Token-time interaction	No
B3	Memory-Augmented Timeformer	Token-time interaction	Causal prototypes

4.1 Static Transformer

The Static Transformer is the baseline encoder. It receives token and position embeddings but no explicit epoch information:

$$x_i = e(w_i) + p_i.$$

The resulting sequence is passed through a small pre-layer-normalized Transformer encoder. This model can learn corpus-level co-occurrence patterns, but it has no direct way to distinguish the representation of a subject at t_0 from the same subject at t_9 , except through the local words observed in the sentence. It therefore represents the null hypothesis that temporal behavior can be recovered from static distributional evidence alone.

4.2 Continuous Time Encoding

The temporally conditioned variants use a continuous encoding of epoch index rather than a learned lookup table. For an epoch $t \in \{0, \dots, T\}$, the index is normalized to $[0, 1]$, mapped to fixed sinusoidal features, and then projected by a small MLP:

$$\tau(t) = \text{MLP}(\text{sinusoidal}(t/T)).$$

This choice is meant to support interpolation and extrapolation across the small number of epochs in the benchmark. A learned embedding table could memorize the training epochs as unrelated identifiers; a continuous encoding instead treats epochs as ordered points on a shared temporal scale. The sinusoidal features here encode epoch index as a scalar on a continuous temporal scale; they are distinct from the token-position encodings used by the Transformer encoder itself, which remain standard positional embeddings.

4.3 Additive Time-Conditioned Transformer

The Additive Time-Conditioned Transformer injects the same epoch vector into every token position:

$$x_i = e(w_i) + p_i + \tau(t).$$

This model tests the minimal effect of exposing the encoder to the current epoch. Because the same temporal offset is added to every token, the model is given global period information but no token-specific temporal modulation at the embedding layer. It therefore isolates the question of whether simply telling the model the epoch is sufficient for traceability.

4.4 Token-Time Transformer

The Token-Time Transformer replaces additive conditioning with an explicit interaction between lexical identity and epoch. For each token, the token embedding and time encoding are concatenated and projected back to the model dimension:

$$x_i = W[e(w_i); \tau(t)] + p_i.$$

Here $e(w_i), \tau(t) \in \mathbb{R}^d$, so the concatenation $[e(w_i); \tau(t)] \in \mathbb{R}^{2d}$ and $W \in \mathbb{R}^{d \times 2d}$ projects back to the model dimension. This is the central Timeformer mechanism in the paper. Unlike the additive variant, it allows the same epoch vector to affect different lexical items in different ways before self-attention is applied. Thus, a subject token can receive a different temporally conditioned representation from a verb or object in the same sentence, and the same subject can be represented differently across epochs. The empirical question is whether this token-time interaction produces more traceable semantic neighborhoods than global time conditioning alone.

4.5 Memory-Augmented Timeformer

The Memory-Augmented Timeformer extends the Token-Time Transformer with a causal prototype memory. For each subject s and epoch t , a prototype $m(s, t)$ is computed as the mean of subject representations observed in training sentences for that subject and epoch. Prototypes are computed with stop-gradient, from training data only, and are served causally: at epoch t_k , the model may attend only to $\{m(s, t) : t < t_k\}$.

The memory module is a temporal cross-attention layer applied only to the subject embedding. The current subject embedding acts as the query, and the historical prototypes for the same subject act as keys and values. The updated subject embedding is then inserted back into the token sequence before the Transformer encoder:

$$\tilde{x}_s = x_s + g \cdot (\text{LN}(x_s + \text{Attn}(x_s, M_s, M_s)) - x_s),$$

where M_s is the causal history for subject s , and $g \in [0, 1]$ is a learned scalar gate initialized to zero so that at the start of training $\tilde{x}_s = x_s$ and the memory path is inactive. The formula interpolates between the original embedding ($g = 0$) and the layer-normalized attended representation ($g = 1$), allowing the model to gradually open the memory channel as training progresses. Applying the memory update before the textual encoder lets the subsequent self-attention layers propagate historical information from the subject to the other token positions, creating a path by which the MLM objective can reward useful memory use.

In this paper, the memory-augmented variant is treated as a diagnostic extension. It tests whether mean-pooled causal history helps temporal traceability beyond token-time interaction. Because a single mean prototype can collapse coexisting senses within an epoch, the model is not expected to solve sense bifurcation by construction; this limitation is part of the motivation for sense-aware memory in future work.

5 Evaluation Protocol

All model variants are trained with the same self-supervised objective: masked language modeling over the synthetic SVO corpus. The planted `true_context` label is never used to train the encoder. It is used only after training, to evaluate whether the learned representation contains the temporal semantic information implied by the benchmark trajectories. This separation is important because the intended use case is representation learning from corpora in which semantic-change labels are not available.

The main object of evaluation is h_s , the final hidden state at the subject position. This choice follows directly from the benchmark design: subjects are the lexical items with planted temporal trajectories, while verbs and objects serve as noisy contextual evidence. We therefore evaluate whether h_s encodes the intended context of the subject at a given epoch. Sentence-level representations

Figure 2. Timeformer architecture detail

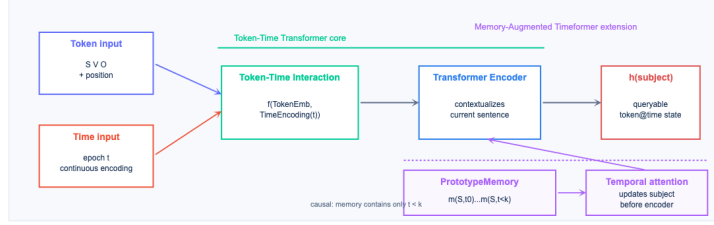


Fig. 2. Architecture detail for the Token-Time Transformer and the optional memory-augmented extension. The memory module attends only to causal historical prototypes and updates the subject representation before Transformer encoding.

based on [CLS] are also computed, but the subject representation is the primary diagnostic target.

We use four complementary diagnostics. First, a linear probe is trained on h_s from the standard test split to predict `true_context` and is then evaluated on each split. This measures whether context information is linearly recoverable from the representation, but it does not by itself establish traceability. Second, we compute a context drift score over nearest neighbors. For each epoch and subject class, we retrieve the k nearest neighbors of each subject representation and measure the proportion assigned to context A, with k fixed at the same value across all models and splits. A traceable representation should show a strong movement from A-like to B-like neighborhoods for drifting subjects, weaker movement for bifurcating subjects, and comparatively limited movement for stable subjects. Third, we use contrastive temporal examples in which the same subject token is presented with two different epoch labels. We define the contrastive sign-flip rate as the proportion of such pairs for which swapping the epoch label reverses the nearest-neighbor context assignment — from A-dominant to B-dominant or vice versa. A model without temporal capacity is expected to produce a sign-flip rate of zero, since the same token yields the same representation regardless of the epoch label provided. Fourth, we evaluate continuation examples from held-out later epochs to test whether temporal behavior extends beyond the training distribution.

The ablation comparisons follow the architecture chain. The gain from time-conditioning is measured by comparing the Additive Time-Conditioned Transformer to the Static Transformer, primarily on the ambiguous split where local markers are uninformative. The gain from token-time interaction is measured by comparing the Token-Time Transformer to the additive variant on the same split. The memory contribution is measured by comparing the Memory-Augmented

Timeformer to the Token-Time Transformer on continuation and contrastive diagnostics.

For the memory-augmented model, we include controls that test whether any gain comes from the correct subject history rather than from extra parameters. In the shuffled-memory control, a subject receives prototypes from a randomly selected different subject, establishing whether subject-specific identity in the causal history matters or whether any historical context produces the same effect. In the no-history control, the memory path is present but receives no valid historical prototypes. We also report an oracle-memory diagnostic, where available, to test whether perfect historical prototypes would be sufficient to improve the same metrics. These controls are interpreted conservatively: memory is considered useful only when it improves over the Token-Time Transformer and over inappropriate or empty histories.

6 Results

We report results over three training runs. Unless otherwise noted, values are means over seeds, with the standard deviation shown in parentheses when space allows. The main finding is that token-time interaction produces a much clearer temporal movement in representation space than the Static Transformer. The memory-augmented variant is informative as a diagnostic extension, but it does not provide a robust improvement over the Token-Time Transformer.

6.1 Temporal Neighborhood Movement

Figure 3 shows the context drift score across epochs. For drifting subjects, the Static Transformer moves only weakly from an A-like neighborhood to a B-like neighborhood: the estimated context-A proportion among nearest neighbors drops from 0.618 at t_0 to 0.387 at t_9 , a change of -0.231 . The Token-Time Transformer shows a much stronger transition, from 0.847 to 0.233, a change of -0.614 . This is the clearest evidence for temporal traceability in the current experiments: the same subject token occupies substantially different neighborhoods when queried at different epochs.

The class-specific curves are also useful as controls. Stable subjects are not perfectly flat, especially for temporally conditioned models, but their movement is much smaller and less aligned with a full A-to-B transition. Across seeds, the stable-class change is $+0.020$ for the Static Transformer, -0.203 for the Token-Time Transformer, and -0.153 for the Memory-Augmented Timeformer. Bifurcating subjects show intermediate behavior: the Token-Time Transformer moves from 0.895 to 0.521, consistent with a trajectory toward a mixed state rather than a complete transition to context B. Thus, the geometric diagnostic distinguishes the qualitative trajectory classes better than a single aggregate probe score.



Fig. 3. Context drift score across epochs. The Token-Time Transformer shows stronger movement from context A toward context B for drifting subjects than the Static Transformer, while stable and bifurcating subjects provide class-specific controls.

Table 2. Context drift score at the first and last epochs. Values are context-A proportions among nearest neighbors; $\Delta = t_9 - t_0$.

Class	Model	t0	t9	Δ
Stable	Static Transformer	0.736	0.756	+0.020
Stable	Token-Time Transformer	0.843	0.641	-0.203
Stable	Memory-Augmented Timeformer	0.810	0.658	-0.153
Drift	Static Transformer	0.618	0.387	-0.231
Drift	Token-Time Transformer	0.847	0.233	-0.614
Drift	Memory-Augmented Timeformer	0.836	0.263	-0.573
Bifurcation	Static Transformer	0.794	0.682	-0.112
Bifurcation	Token-Time Transformer	0.895	0.521	-0.374
Bifurcation	Memory-Augmented Timeformer	0.905	0.517	-0.388

6.2 Probe and Contrastive Diagnostics

The linear probe results are supportive but less decisive than the neighborhood analysis. On the ambiguous split, where local verb and object markers are uninformative, the Static Transformer reaches 0.577 accuracy, the Additive Time-Conditioned Transformer reaches 0.573, and the Token-Time Transformer reaches 0.599. The corresponding mean delta for token-time interaction is +0.026, but it is seed-sensitive: the three runs give +0.053, +0.005, and +0.021. This confirms that probe accuracy alone should not carry the central claim. It measures recoverability of the planted label, not the geometry of temporal movement.

The contrastive sign-flip diagnostic gives a different view. The Static Transformer has a sign-flip rate of 0.000, as expected for a model without explicit time. The Additive Time-Conditioned Transformer averages 0.690, while the Token-Time Transformer averages 0.517. This ordering may appear counterintuitive given that the Token-Time Transformer produces a much stronger neighborhood drift. The decoupling arises because sign-flip rate and drift score measure distinct properties: sign-flip counts the proportion of contrastive pairs in which the nearest-neighbor context assignment crosses the A/B midpoint, while drift score measures the full magnitude of trajectory movement across epochs. A global additive epoch offset displaces all token representations uniformly, which can push many assignments across the decision boundary without producing coherent subject-level trajectories. Token-time interaction modulates each token according to its identity, producing more structured and subject-specific movement — evidenced by the stronger drift — but not necessarily crossing the contrastive threshold in more pairs. We therefore treat contrastive sign flip as evidence that time conditioning changes model behavior at all, and the neighborhood trajectories of Figure 3 as the more diagnostic measure of temporal traceability.

Table 3. Main traceability diagnostics. Accuracy columns report linear-probe accuracy on h_s ; flip is contrastive sign-flip rate. A dash indicates the metric was not computed for this variant.

Model	Drift	Flip	Ambig.	Cont.
Static Transformer	-0.231	0.000	0.577	0.719
Additive Time-Conditioned Transformer	—	0.690	0.573	0.754
Token-Time Transformer	-0.614	0.517	0.599	0.766
Memory-Augmented Time-former	-0.573	0.506	0.599	0.766

6.3 Memory Diagnostics

The Memory-Augmented Timeformer does not produce a robust gain over the Token-Time Transformer. On continuation, the mean accuracy of the Token-Time Transformer is 0.766, and the mean accuracy of the Memory-Augmented

Timeformer is also 0.766. The memory delta across the three runs is effectively zero on average, with seed-level values $+0.013$, $+0.005$, and -0.018 . This result argues against presenting mean-pooled prototype memory as the main empirical contribution.

The oracle-memory diagnostic further clarifies the failure mode. In the corrected oracle run, learned memory slightly improves continuation over its Token-Time reference (0.775 vs. 0.762), while oracle memory does not (0.754). However, oracle memory improves contrastive sign flip relative to the same reference (0.586 vs. 0.448). The careful interpretation is that perfect historical prototypes can affect contrastive temporal behavior, but historical memory alone does not solve the continuation objective under the current MLM setup. This supports the use of B3 as a diagnostic extension and motivates memory mechanisms that preserve multiple senses rather than averaging them into a single prototype.

Table 4. Memory diagnostics from the corrected oracle run. Continuation is linear-probe accuracy on held-out later epochs; flip is contrastive sign-flip rate.

Variant	Continuation	Sign flip	Delta vs. B2b ref.
B2b reference	0.762	0.448	–
B3 learned	0.775	0.448	$+0.013$ cont.; $+0.000$ flip
B3 oracle	0.754	0.586	-0.008 cont.; $+0.138$ flip

7 Discussion

The results support a clear conclusion: temporal traceability is a geometric property of the representation that does not reduce to predictive accuracy. The Token-Time Transformer improves probe accuracy on the ambiguous split by only $+0.026$ on average — a modest and seed-sensitive gain. Yet the same model shows a neighborhood drift of -0.614 for drifting subjects, compared to -0.231 for the Static Transformer — a difference that is stable across seeds and visible in the class-specific trajectory curves of Figure 3. This dissociation is the central empirical finding: a model can condition its outputs on epoch information without exposing a representation in which the same lexical item occupies meaningfully different neighborhoods at different epochs. The converse also holds: a model with explicit token-time interaction can produce coherent geometric movement even when the improvement in probe accuracy is modest. Evaluating traceability therefore requires diagnostics that inspect the geometry of the representation directly, not only those that measure label recoverability.

A linear probe trained on h_s establishes whether the planted context label is linearly recoverable from the subject representation. A high probe score is consistent with traceability, but it does not establish it: the probe could be recovering a global epoch signal distributed across the representation rather than a

subject-specific temporal trajectory. The ambiguous split is the most informative probe condition precisely because local verb and object markers are uninformative there, forcing any gain to come from the interaction between subject identity and epoch. Even in this condition, differences across model variants are small and unstable across seeds. This does not mean the probe is uninformative — it provides a lower bound on temporal information encoded in the representation — but it is not the appropriate primary evidence for a traceability claim. The nearest-neighbor diagnostics, which directly inspect where each subject representation sits relative to its temporal context, are more suitable for that purpose.

The additive variant demonstrates that exposing the encoder to the epoch is not sufficient for traceability. A global additive epoch offset shifts all token representations by the same vector, leaving the relative geometry of subject representations at different epochs only weakly constrained. The model knows which period it is processing, but this knowledge does not translate into subject-specific trajectory structure — as confirmed by the high sign-flip rate without a corresponding coherent drift. Token-time interaction changes this fundamentally: by concatenating each token embedding with the epoch encoding before projection, the model allows the same epoch vector to affect different tokens differently. The same epoch produces a distinct transformation for the subject token and for the verb or object in the same sentence, and the same subject receives a different embedding at different epochs. This is the mechanism behind the coherent neighborhood trajectories in the drift analysis: the representation of each subject is a function of both its identity and its epoch, not merely of the local context words observed in the sentence.

The Memory-Augmented Timeformer tests the simplest form of causal memory: a mean of historical subject representations per epoch. The oracle diagnostic clarifies the failure mode. Perfect historical prototypes improve the contrastive sign-flip rate but not continuation accuracy, and learned prototypes produce near-zero average gain over the Token-Time Transformer. The limitation is structural: a single mean prototype per subject per epoch collapses the internal distribution of that subject’s representations at that time. For drifting subjects, where all representations in an epoch are drawn from the same dominant context, the mean captures the trajectory reasonably well. For bifurcating subjects, which develop two coexisting senses in later epochs, the mean is a centroid between two clusters — a point that represents neither sense well. This is not a failure of the memory architecture, but a failure of the aggregation function. Useful causal memory for traceable representations should preserve the internal structure of historical representations — for example, through mixture models, exemplar-based attention, or clustering over historical states. This limitation is the direct motivation for the sense-aware memory mechanism studied in the companion paper.

8 Conclusion

This paper studied whether a Transformer-style encoder can expose a temporally traceable representation — one in which the same lexical item, queried at different epochs, occupies neighborhoods that reflect its intended temporal context. The answer, in the controlled benchmark used here, is conditional: explicit token-time interaction is necessary. A static encoder and a globally time-conditioned encoder both fail to produce coherent subject-level trajectories. An encoder that conditions each token embedding on the epoch before self-attention is applied does produce such trajectories, with a neighborhood drift for drifting subjects that is more than twice that of the static baseline (-0.614 vs. -0.231).

The contribution is narrow and deliberate. Token-time interaction supports temporal traceability in a controlled diagnostic setting; it is not a claim about open-domain historical language or about the superiority of any particular architecture. The evaluation protocol — combining nearest-neighbor drift, contrastive sign-flip rate, probe-based accuracy, and continuation diagnostics — also contributes a method for separating representational traceability from predictive performance, a distinction the results show to be empirically real.

The memory-augmented variant clarifies what remains unsolved. Mean-pooled historical prototypes do not produce a robust improvement over token-time interaction, and the oracle diagnostic identifies the reason: single-prototype aggregation collapses coexisting senses that bifurcating subjects develop in later epochs. Sense-aware memory mechanisms — capable of preserving distributional structure rather than averaging it away — are the natural next step and are the subject of the companion paper.

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